Ex. No. 2: Implementing TDMA Slot Allocation in GSM.

AIM:

To implement TDMA (Time Division Multiple Access) slot allocation in GSM using MATLAB and analyze system performance in terms of time slot duration, frame utilization, and number of allocated slots.

Objective:

1. To calculate the Time Slot Duration in a GSM TDMA frame by dividing the total frame time into equal time slots. This helps in understanding the time structure of GSM communication.
2. To determine the Frame Utilization (%) based on the number of allocated time slots compared to total available slots. This evaluates the efficiency of TDMA resource usage.
3. To compute the Number of Allocated Time Slots for multiple users in the GSM system. This ensures proper scheduling and interference-free communication.

Objective 1: Matlab Code :

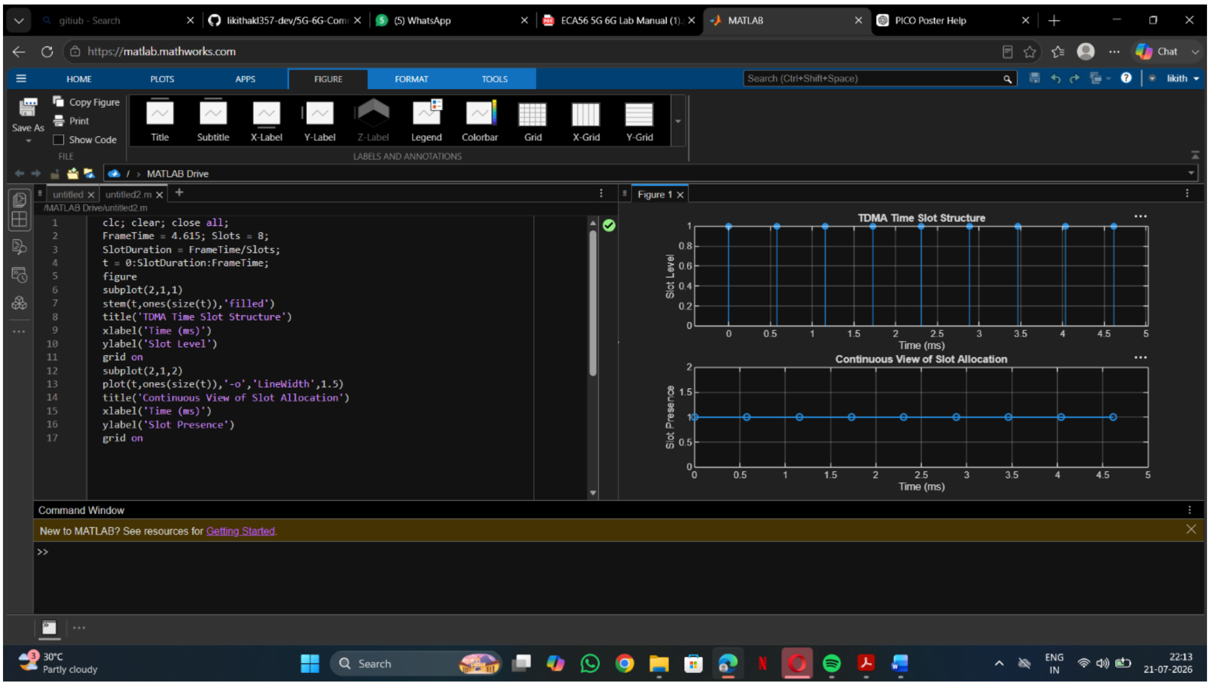
```
clc; clear; close all;

% TDMA Parameters

FrameTime = 4.615; % Frame Duration (ms)

Slots = 8; % Number of Time Slots

SlotDuration = FrameTime / Slots;

% Time Vector

t = 0:SlotDuration:FrameTime;

% Plot TDMA Slot Structure figure;

% Discrete Time Slots

subplot(2,1,1);

stem(t, ones(size(t)), 'filled');

title('TDMA Time Slot Structure');

xlabel('Time (ms)');

ylabel('Slot Level');

grid on;

% Continuous Slot Allocation

subplot(2,1,2);

plot(t, ones(size(t)), '-o', 'LineWidth', 1.5);

title('Continuous View of Slot Allocation');
```

xlabel('Time (ms)');

ylabel('Slot Presence');

grid on;

Output Objective 1:



Objective 2: Matlab Code :

clc; clear; close all;

% GSM Frame Parameters

TotalSlots = 8;   % Total Time Slots

AllocatedSlots = 6;   % Allocated Slots

FreeSlots = TotalSlots - AllocatedSlots;

% Frame Utilization

Utilization = (AllocatedSlots / TotalSlots) * 100;

% Display Utilization

disp('Frame Utilization (%)');

disp(Utilization);

% Plot Results

figure;

% Graph 1: Line Plot

subplot(2,1,1);

plot([1 2], [AllocatedSlots FreeSlots], '-o', 'LineWidth', 2);

title('GSM Frame Utilization (Allocated vs Free)');

xlabel('Slot Type Index (1 = Allocated, 2 = Free)');

ylabel('Number of Slots');

grid on;

% Graph 2: Pie Chart

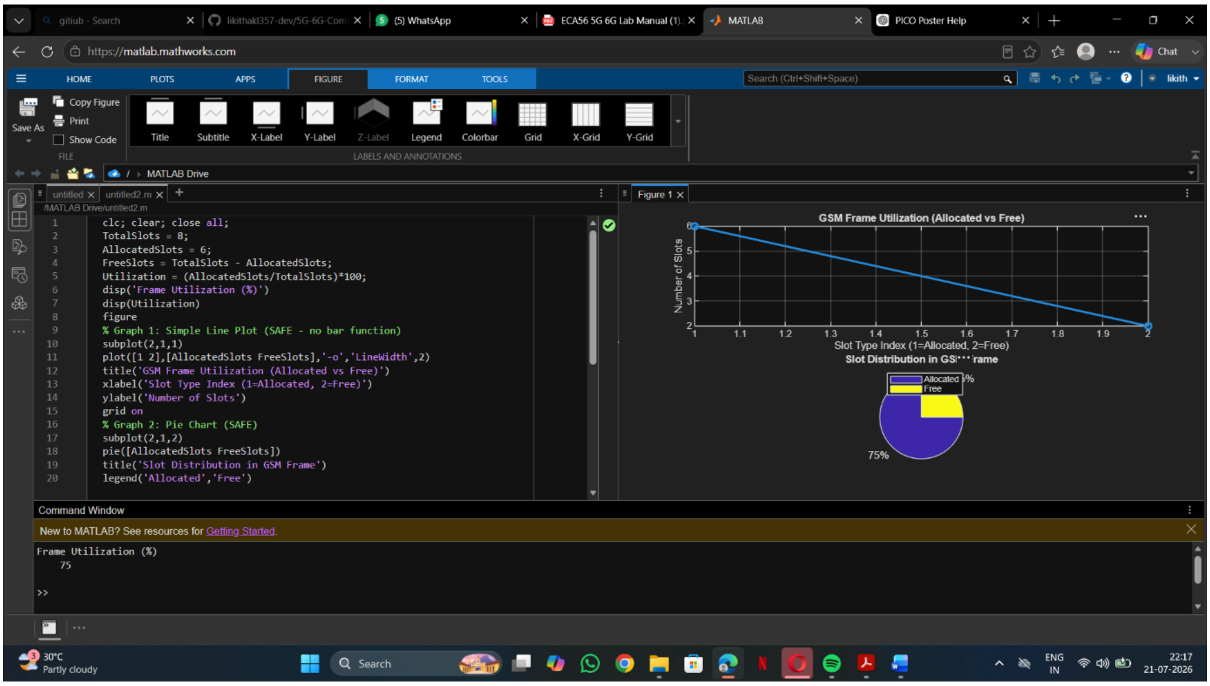
subplot(2,1,2);

pie([AllocatedSlots FreeSlots]);

title('Slot Distribution in GSM Frame');

legend('Allocated', 'Free');

Output Objective 2:



Objective 3: Matlab Code:

clc;

clear;

close all;

% TDMA Parameters

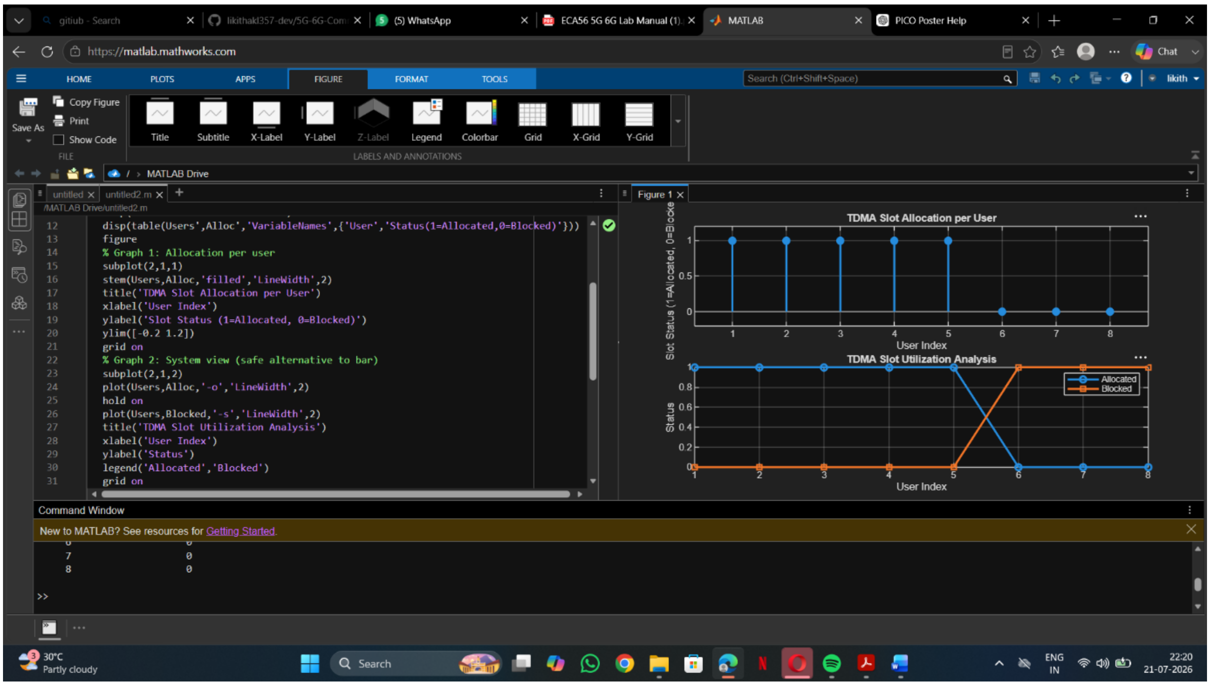
Users = 1:8;   % Total Users

TotalSlots = 5;   % Available Time Slots

% Initialize Slot Allocation

```
Alloc = zeros(1,8);
Alloc(1:TotalSlots) = 1;  % Assign slots to first users
Blocked = 1 - Alloc;      % Remaining users are blocked
% Display Allocation Status
disp('User Wise Slot Allocation');
disp(table(Users', Alloc', ...
    'VariableNames', {'User', 'Status(1=Allocated,0=Blocked)'}));
% Plot Results
figure;
% Graph 1: Slot Allocation per User
subplot(2,1,1);
stem(Users, Alloc, 'filled', 'LineWidth', 2);
title('TDMA Slot Allocation per User');
xlabel('User Index');
ylabel('Slot Status (1 = Allocated, 0 = Blocked)');
ylim([-0.2 1.2]);
grid on;
% Graph 2: Slot Utilization Analysis
subplot(2,1,2);
plot(Users, Alloc, '-o', 'LineWidth', 2);
hold on;
plot(Users, Blocked, '-s', 'LineWidth', 2);
title('TDMA Slot Utilization Analysis');
xlabel('User Index');
ylabel('Status');
legend('Allocated', 'Blocked');
grid on;
```

Output Objective 3 :



Result:

1. The time slot duration in the GSM TDMA frame was successfully calculated, confirming equal division of the frame into fixed time intervals.
2. The frame utilization was effectively analyzed, showing the proportion of allocated and free slots in the TDMA system.
3. The slot allocation per user was successfully implemented, demonstrating that limited slots are assigned to users while excess users are blocked due to resource constraints.